QUASI-COHERENCE AND COHOMOLOGICAL VANISHING OF THE DERIVED PUSHFORWARD ALONG A QUASI-COMPACT QUASI-SEPARATED MORPHISM

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Allowed standard prerequisites (editorial).

- For a quasi-compact quasi-separated morphism there is a uniform bound N with R^p f_* vanishing on quasi-coherent modules in degrees p >= N, and the vanishing persists after base change, by the cohomology-of-schemes quasi-coherence lemma. invoked at proof lines 682-685. Supplies the integer N used throughout the proof.
- The Grothendieck spectral sequence R^p f_* H^q(E) => R^{p+q} f_* E for the composition of derived functors. invoked at proof lines 691-694. Handles the bounded-below case of parts (1), (2), (3).
- The K-injective computation R Gamma(U, Rf_* E) = R Gamma(f^{-1}(U), E) on affine opens of the base, valid for the section functor by the nice-K-injective application lemma. invoked at proof lines 705-709. Reduces the vanishing assertions (2) and (3) to cohomology of sections over affine opens.
- The truncation distinguished triangle tau_{<= -n-1} E -> E -> tau_{>= -n} E -> (tau_{<= -n-1} E)[1] in the derived category. invoked at proof lines 716-722. Extends part (1) from bounded-below to arbitrary E by reducing each cohomology degree to a bounded-below truncation.

Source reference. Original lemma 01XJ: sources/stacks-chow/tags.txt, source lines 2520–2520. Cited at proof lines 682-683 to pick the bound N; the choice and its use are inside the counted span.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1510–1510. Cited at proof lines 694-695 for the spectral sequence in the bounded-below case; the case analysis is inside the counted span.

Source reference. Original lemma 01XK: sources/stacks-chow/tags.txt, source lines 2521–2521. Cited at proof lines 700-702 to get vanishing of H^p(f^{-1}(U), F); the vanishing argument is inside the counted span.

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Source reference. Original lemma 08D4: sources/stacks-perfect/pinned-original-source.tex, source lines 375–375. Cited at proof lines 705-706 for R Gamma(U, Rf_* E) = R Gamma(f^{-1}(U), E); the vanishing argument is inside the counted span.

Source reference. Original remark 08J5: sources/stacks-chow/tags.txt, source lines 10838–10838. Cited at proof lines 721-722 for the triangle used in the general case of part (1); the reduction argument is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume that f is quasi-separated and quasi-compact.*

1. *The functor Rf_* sends $D_{QCoh}(\mathcal{O}_X)$ into $D_{QCoh}(\mathcal{O}_S)$.*
2. *If S is quasi-compact, there exists an integer $N = N(X, S, f)$ such that for an object E of $D_{QCoh}(\mathcal{O}_X)$ with $H^m(E) = 0$ for $m > 0$ we have $H^m(Rf_*E) = 0$ for $m \geq N$.*
3. *In fact, if S is quasi-compact we can find $N = N(X, S, f)$ such that for every morphism of schemes $S' \to S$ the same conclusion holds for the functor $R(f')_*$ where $f' : X' \to S'$ is the base change of f.*

Proof. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. To prove (1) we have to show that Rf_*E has quasi-coherent cohomology sheaves. The question is local on S, hence we may assume S is quasi-compact. Pick $N = N(X, S, f)$ as in Cohomology of Schemes, Lemma 01XJ. Thus $R^p f_*\mathcal{F} = 0$ for all quasi-coherent $\mathcal{O}_X$-modules $\mathcal{F}$ and all $p \geq N$ and the same remains true after base change.

First, assume E is bounded below. We will show (1) and (2) and (3) hold for such E with our choice of N. In this case we can for example use the spectral sequence

$$R^p f_* H^q(E) \Rightarrow R^{p+q} f_* E$$

(Derived Categories, Lemma 015J), the quasi-coherence of $R^p f_* H^q(E)$, and the vanishing of $R^p f_* H^q(E)$ for $p \geq N$ to see that (1), (2), and (3) hold in this case.

Next we prove (2) and (3). Say $H^m(E) = 0$ for $m > 0$. Let $U \subset S$ be affine open. By Cohomology of Schemes, Lemma 01XK and our choice of N we have $H^p(f^{-1}(U), \mathcal{F}) = 0$ for $p \geq N$ and any quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$. Hence we may apply Lemma 08D4 to the functor $\Gamma(f^{-1}(U), -)$ to see that

$$R\Gamma(U, Rf_*E) = R\Gamma(f^{-1}(U), E)$$

has vanishing cohomology in degrees $\geq N$. Since this holds for all $U \subset S$ affine open we conclude that $H^m(Rf_*E) = 0$ for $m \geq N$.

Next, we prove (1) in the general case. Recall that there is a distinguished triangle

$$\tau_{\leq -n-1}E \to E \to \tau_{\geq -n}E \to (\tau_{\leq -n-1}E)[1]$$

in $D(\mathcal{O}_X)$, see Derived Categories, Remark 08J5. By (2) we see that $Rf_*\tau_{\leq -n-1}E$ has vanishing cohomology sheaves in degrees $\geq -n+N$. Thus, given an integer q we see that $R^q f_*E$ is equal to $R^q f_*\tau_{\geq -n}E$ for some n and the result above applies. $\square$